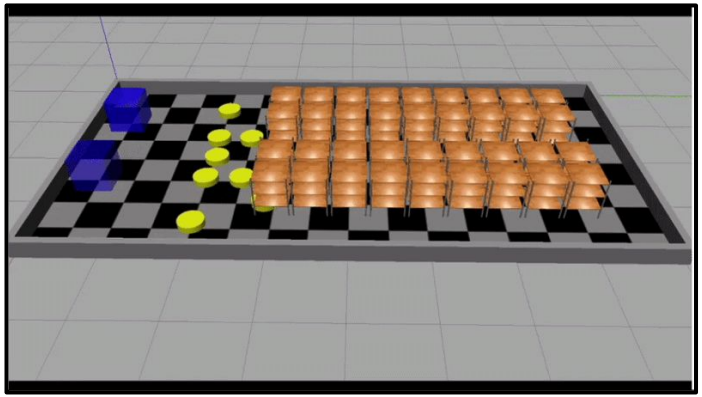
# Model-Based Diffusion Optimal Control for Multi-Robot Motion Planning

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https://github.com/hhhhzl/mdoc

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# Motivations



Multi-Robot Motion Planning (MRMP) **requires**:

- Dynamic feasibility
- Collision-free trajectories
- Scalability

Why the problem is **hard**?

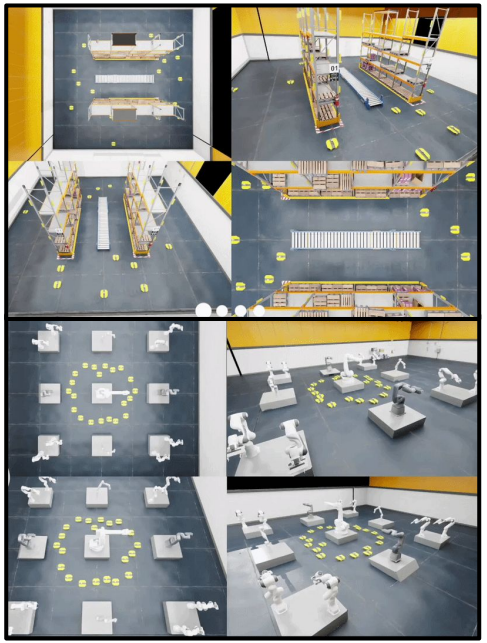


- High dimensional space (joint trajectory space)
  - Classical sampling-based methods are **slow and sample-inefficient**.
  - Optimization-based methods require solving **large non-convex constrained programs.**
- Coupled safety constraints
  - Dynamics, obstacles, and robot-robot collisions **must be satisfied** simultaneously.

# Motivation

Multi-Robot Motion Planning with Diffusion Models (MMD) [Shaoul et al. 2025]

- Learning-based
- Strength in generating multi modality trajectory solutions
- Issues?
  - **Need pre-generated or pre-collected data  (massive!)**
  - **Pretrain a network (model-free)**
  - **Dynamics?? Constraints??**

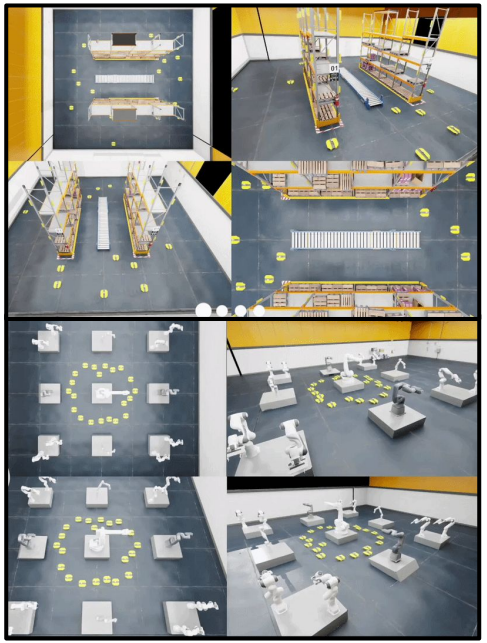


Yorai Shaoul, Itamar Mishani, Shivam Vats, Jiaoyang Li, and Maxim Likhachev. Multi-robot motion planning with diffusion models. In International Conference on Learning Representations, 2025.

# Motivation

Multi-Robot Motion Planning with Diffusion Models (MMD) [Shaoul et al. 2025]

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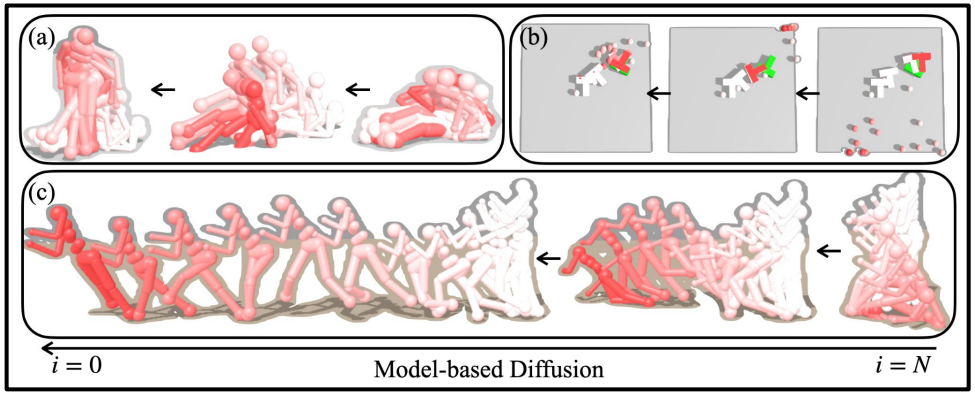
**We Need Model-Based Approach!!**

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# Motivation (Intro)

Model-Based Diffusion (MBD) as Trajectory Optimization for Motion Planning [Pan et al. 2024]

$$\tau^\star \in \arg\min_{\tau^{1:T}} \quad \mathcal{J}(\tau^{1:T})$$
$$\text{s.t.} \quad s^{t+1} = f(s^t, u^t), \quad g^t(s^t, u^t) \leq 0,$$




Chaoyi Pan, Zeji Yi, Guanya Shi, and Guannan Qu. Model-based diffusion for trajectory optimization. In Advances in Neural Information Processing Systems, pages 57914–57943, 2024.

# Motivation (Intro)

Diffusion Steps in Loop

Sample From Gaussian Kernel

$$\tilde{\tau} \sim \Omega_k := \mathcal{N}\left(\frac{\tau_k}{\sqrt{\bar{\alpha}_k}}, \left(\frac{1}{\bar{\alpha}_k} - 1\right)\mathbf{I}\right),$$

Sampling on control input

Rollout Dynamics

$$s^{t+1} = f(s^t, u^t),$$

Getting Batch Nominal Trajectories

Monte Carlo Reweighted

$$\bar{\tau}_k = \frac{\sum_{m=1}^{M} p_1(\tilde{\tau}_{k,m})\,\tilde{\tau}_{k,m}}{\sum_{m=1}^{M} p_1(\tilde{\tau}_{k,m})},$$

Evaluated based on cost or reward

Diffusion Score Compute

$$\hat{\mathbf{S}}_k \approx -\frac{\tau_k}{1-\bar{\alpha}_k} + \frac{\sqrt{\bar{\alpha}_k}}{1-\bar{\alpha}_k}\,\bar{\tau}_k.$$

Score is analytically calculated !!!

Score = Gradient of Log Density (learned in model-free approaches)

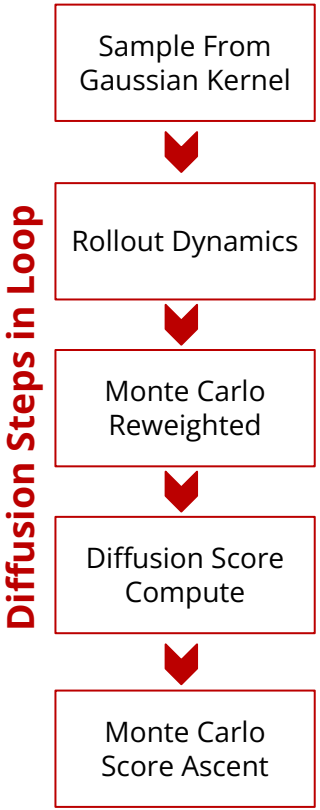
Monte Carlo Score Ascent

$${}^{k-1}\tau^{1:H} = \frac{1}{\sqrt{\alpha_k}}\left({}^{k}\tau^{1:H} + (1-\bar{\alpha}_k)\,\hat{\mathbf{S}}_k\right),$$

Trajectory update

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# Motivation (Intro)




Vanila-MBD

- Issues?
  - **Score are not constraint-aware**
  - **Plan for single robot trajectory**

**Thus, We need it lift to**

**(1) Constrained robot planning**

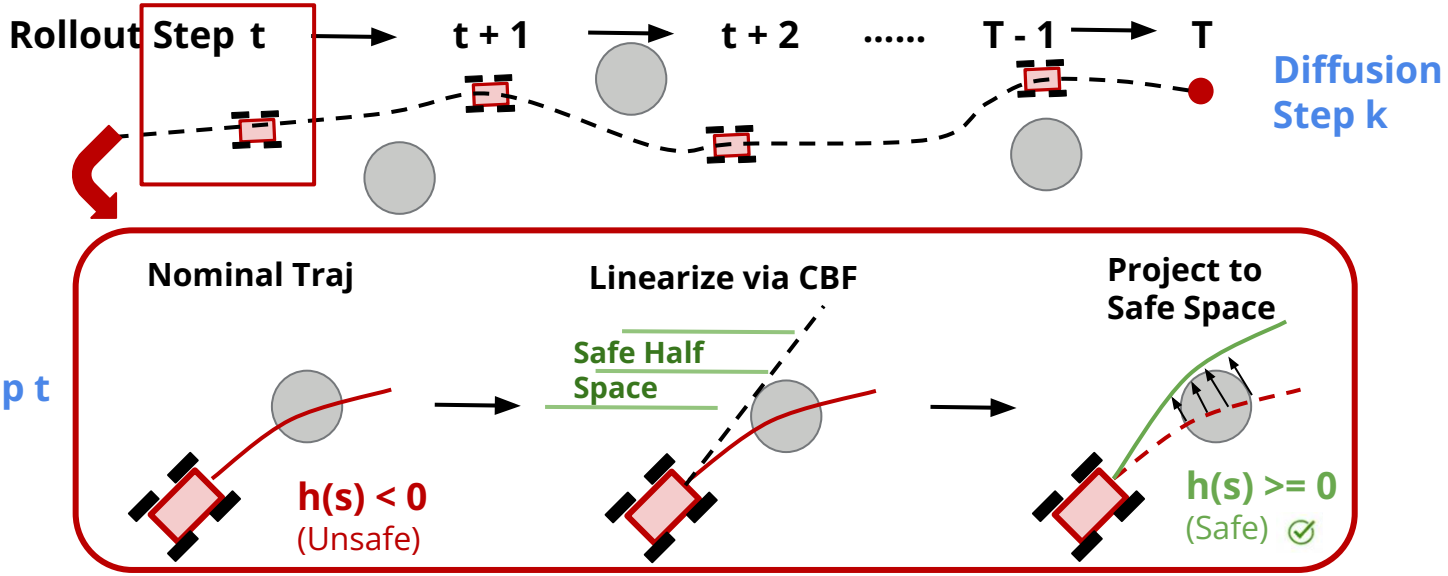
**(2) Multi-robot motion planning**

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# MDOC: Single robot planner that handles constraints

Diffusion Steps in Loop

- Sample From Gaussian Kernel
- Rollout Dynamics with Per-Step CBF
- Monte Carlo Reweighted
- Diffusion Score Compute
- Monte Carlo Score Ascent

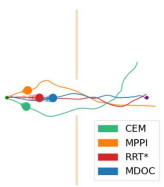


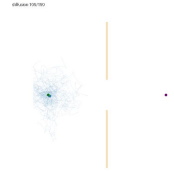

**Safety is implemented during the sampling, before the score is computed.**
**Therefore, the score is constraints-aware.**

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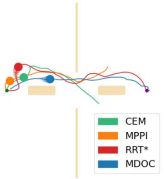
# MDOC: Results

Narrow map (Top)



More-Constrained Narrow map (Bottom)

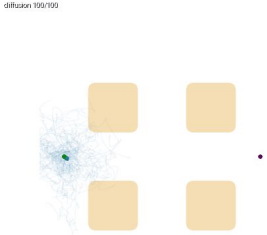


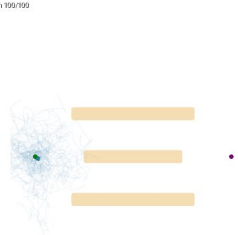


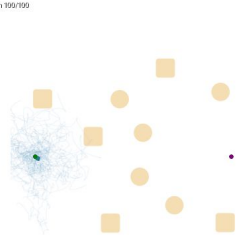

| | CEM | MPPI | RRT* | MDOC |
|---|---|---|---|---|
| Top | $2.1 \pm 0.2$ | $3.2 \pm 0.3$ | $66.1 \pm 6.0$ | **100.0** |
| Bottom | $0.2 \pm 0.1$ | $0.4 \pm 0.1$ | $41.7 \pm 3.4$ | **100.0** |

**100% Sample Trajectories Success Rate**

Additional Results:







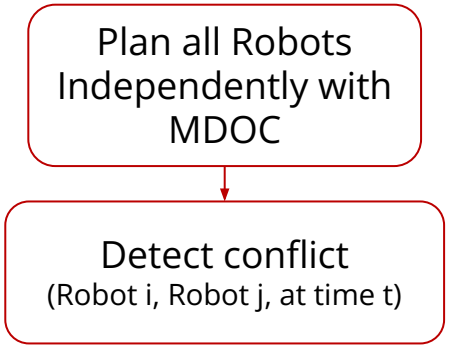
**Efficient in finding feasible path while remaining collision-free in constrained environment**

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# MDOC-CBS: Scalable Multi-robot Coordination with MDOC as the low-level planner

**High-Level Idea:**

- Detect the inter-robot conflicts via **Conflict-Based Search (CBS) [Sharon et al. 2015]**

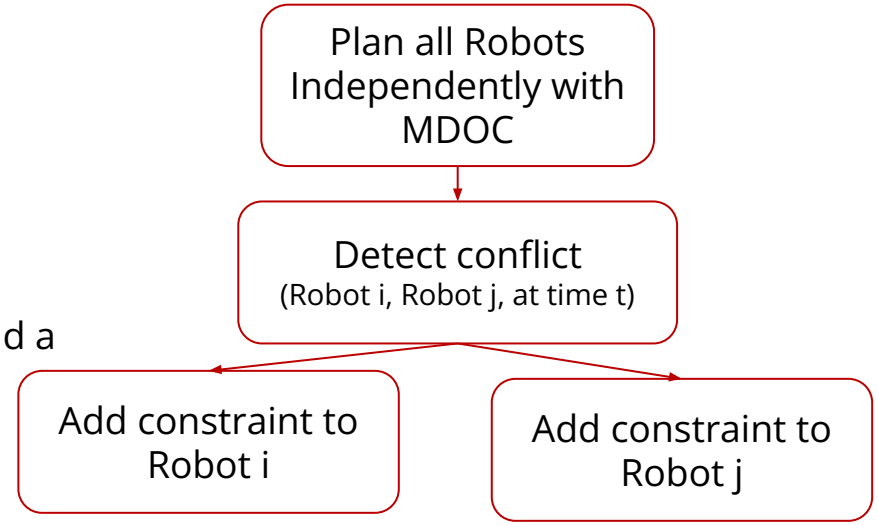



Guni Sharon, Roni Stern, Ariel Felner, and Nathan R Sturtevant. Conflict-based search for optimal multi-agent pathfinding. Artificial intelligence, 219:40–66, 2015.

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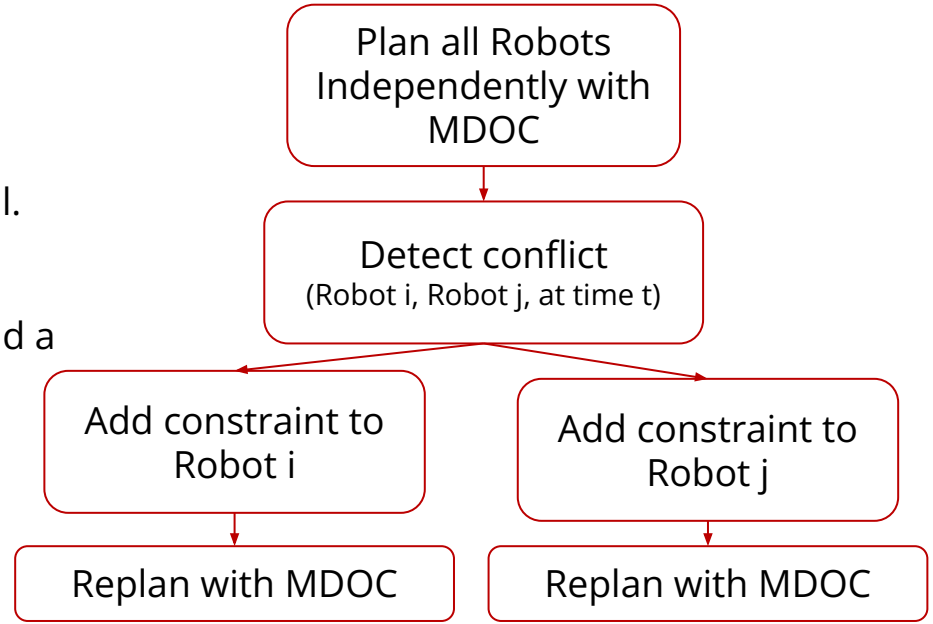
- Detect the inter-robot conflicts via Conflict-Based Search (CBS) [Sharon et al. 2015]
- Expand to a **Constraint Tree (CT)** and add a **time-indexed sphere constraint** to one conflicted robot



# MDOC-CBS: Scalable Multi-robot Coordination with MDOC as the low-level planner

**High-Level Idea:**

- Detect the inter-robot conflicts via Conflict-Based Search (CBS) [Sharon et al. 2015]
- Expand to a Constraint Tree (CT) and add a time-indexed sphere constraint to one conflicted robot
- **Replan such robot** for both inter-robot conflicts and obstacle constraints **with MDOC**

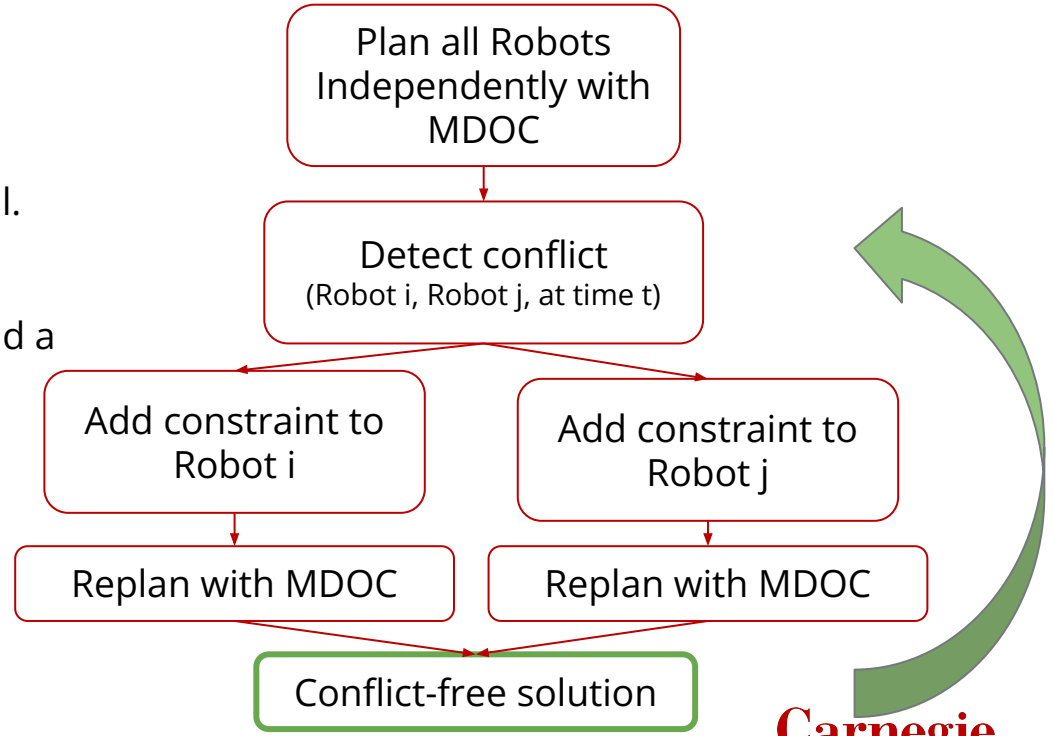



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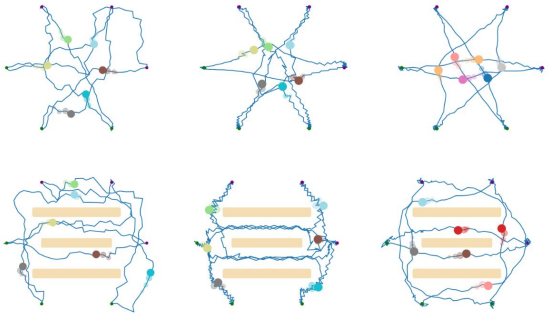
- Detect the inter-robot conflicts via Conflict-Based Search (CBS) [Sharon et al. 2015]
- Expand to a Constraint Tree (CT) and add a time-indexed sphere constraint to one conflicted robot
- Replan such robot for both inter-robot conflicts and obstacle constraints with MDOC
- Repeat until **conflict-free**

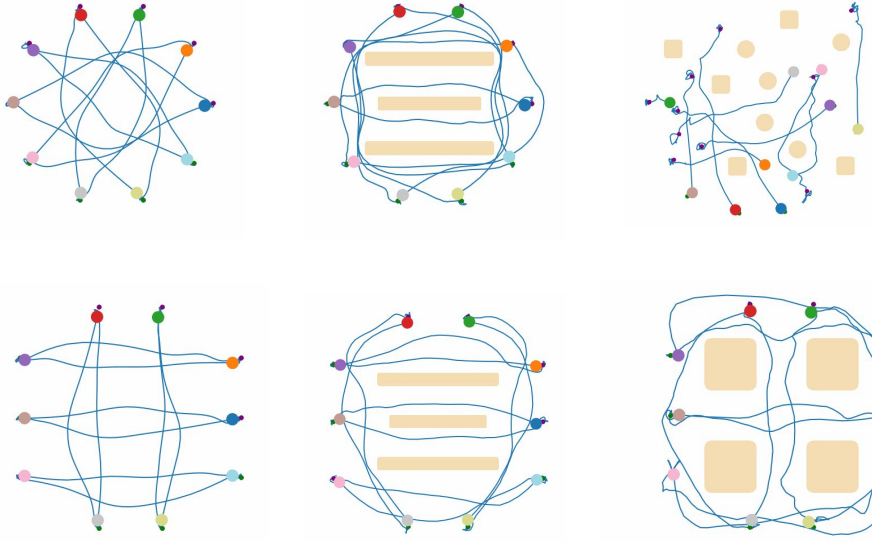



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# MDOC-CBS: Results

KCBS MMD-CBS MDOC-CBS





| Map | Method | P↓ | G (1e-4)↓ |
|---|---|---|---|
| Empty | KCBS | 2.2 ± 0.07 | 20.5 ± 2.75 |
| | MMD-CBS | 2.9 ± 0.06 | 60.6 ± 5.33 |
| | MDOC-CBS | **1.8** ± **0.03** | **6.7** ± **0.05** |
| Conveyor | KCBS | **2.7** ± **0.10** | 20.7 ± 2.12 |
| | MMD-CBS | 3.9 ± 0.11 | 85.0 ± 5.45 |
| | MDOC-CBS | 2.8 ± 0.04 | **12.9** ± **1.40** |

TABLE II: Comparison of methods in terms of average path length **P** and geometric (Laplacian) smoothness **G** over success trails of all agent numbers, evaluated on the Empty map and Conveyor map in the Circle setup.

Key Findings

- Higher Success Rate
- **5x – 10x** lower planning time than MMD-CBS
- Shorter and **smoother** Trajectories
- **Scale to 20 robots and 40 robots in larger maps**

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# Takeaways

- **Training-Free** (for **Different Environments without Re-training**)
- **Low level single-robot planner (MDOC)** that solves both **Conflicts** and **Obstacle Constraints**
  - **Perstep CBF rollout within Dynamics** (Safe and **Fast**)
  - **Diffusion Score is constraints-award** (Not a **Post-hoc Projection**)
- **High level multi-robot planner (CBS)** that detect **Inter-robot Conflicts,** then replan by MDOC

# Thanks